

Project K — B2B Platform

Statement of Work (SOW)

1. Overview

This document defines the production-ready requirements for Project K, a next-generation AI-powered eLearning platform designed for higher education institutions across Tamil Nadu. The system transforms traditional static academic content into AI-driven learning experiences and includes an integrated AI Interview module.

The platform is built on a B2B institutional model. Knotopian (Super Admin) manages affiliated universities and colleges. Each college is administered by Admin & HODs/Faculty. Students access learning content, AI tutoring, assessments, and interview preparation through a dedicated student portal.

2. Vision

To become the primary AI-powered academic companion for every college student in Tamil Nadu, transforming raw academic content into fully structured, ai-driven learning experiences while enabling on-demand personalised tutoring and interview readiness.

3. Product Scope

The platform encompasses the following functional domains:

- Onboarding of affiliated universities and their member colleges
- Super Admin Portal — content upload, AI course generation, university/college management
- College Admin Portal — content upload, AI course generation, multi-role access for admin, HOD and Faculty/SME.
- Student Portal — Classroom (AI video lectures) + QnA AI Tutor + Assessments
- Content-to-Learning Automation: raw academic documents to AI video lessons
- English as the primary instruction language; Tamil language support for QnA interactions
- Desktop web browser as the primary delivery platform
- AI-generated MCQ knowledge checks with auto-evaluation
- Analytics dashboards for Super Admin and College Admin
- Student leaderboard

- AI Interview Module — embedded within the B2B Student Portal (also available as standalone B2C product)

4. Business Model

The platform operates on a B2B institutional licensing model. Knotopian contracts directly with universities and colleges. The platform is not consumer-facing from a billing perspective, institutions pay Knotopian for platform access, and students access the platform using credentials provisioned by their institution.

The embedded AI Interview module within the Student Portal is available to B2B students (**Final year Students only**) without separate subscription. The standalone B2C Interview Portal operates independently with its own subscription model (covered in a separate SOW).

5. User Roles and Portals

5.1 Super Admin (Knotopian)

- Single, platform-wide administrator account
- Manages all universities, colleges, and users across the platform
- Uploads and manages academic content; triggers AI course generation
- Views platform-wide reports and monitors system health
- Accesses aggregated analytics: total institutions, active students, completion rates, average assessment scores, top QnA queries, content engagement metrics (watch time)

5.2 College Admin — HOD / Faculty (SME)

- Creates and manages academic hierarchy (Department, Program, Semester, Subject, Chapter)
- Uploads raw academic content (PDFs, lecture notes, textbooks, question banks) and triggers AI course generation
- Adds and manages students: single add or bulk import via CSV (up to 100 students)
- Assigns students to programs and semesters
- Views and downloads complete student / cohort reports.
- Reviews and approves pre-recorded AI-generated video modules.
- Reviews and approves AI-generated MCQs before student publication.
- Creates and manages interview assignments for student cohorts (B2B Interview-Final year students)
- Faculty role is restricted from accessing institution-wide dashboards (RBAC enforced)

5.3 Student

- Accesses assigned programs, semesters, and subjects
- Watches AI -driven video lectures (Classroom)
- Interacts with the live QnA AI Tutor for doubt resolution
- Takes AI-generated MCQ assessments and downloads PDF reports post-submission
- Views leaderboard ranking after assessment submission
- Final year students takes AI mock interview sessions (voice with avatar interviewer)
- Views interview score report, per-question feedback, and improvement plan

6. Detailed Feature Specification

6.1 Content Upload and AI Course Generation Pipeline

The core content pipeline converts raw academic materials into structured AI-driven learning content.

6.1.1 Content Ingestion

- Supported input formats: PDF, Text files, question banks, PPTs.
- Content is tagged to a specific Chapter within the academic hierarchy
- Content upload is performed either by Super Admin or authorised Faculty/SME

6.1.2 RAG (Retrieval-Augmented Generation) Pipeline

- End-to-end RAG pipeline converts uploaded documents into a structured knowledge base
- Vector embeddings stored in Pinecone or Weaviate for semantic retrieval
- Knowledge base is chapter-scoped to ensure content relevance during generation

6.1.3 AI Avatar Video Course Generation

- System generates an AI video course per chapter, using the chapter knowledge base.
- Video generation is asynchronous, Faculty/SME is notified on completion.
- Generated modules undergo SME review and approval before student publication
- English is the primary language for video instruction.

6.1.4 MCQ Generation

- AI generates multiple-choice questions from chapter content
- MCQs go through a Subject Matter Expert review and approval workflow before being activated for student assessments
- Approved MCQs are tagged to subject and chapter for structured delivery

6.2 Super Admin Portal

- Create and manage University hierarchy: University > College > Department > Program > Semester > Subject > Chapter
- Onboard affiliated universities and member colleges
- Upload academic content and trigger AI course generation pipeline
- Monitor platform health: system status, uptime, error logs
- View platform-wide analytics dashboard (see Section 6.7)
- Manage all users across institutions

6.3 College Admin Portal

Accessible by two sub-roles: Admin , HOD (full college-level access) and Faculty/SME (subject-level access, no institution-wide dashboard).

- Create academic hierarchy below university level
- Upload raw content and trigger AI course generation
- Bulk student import via CSV (up to 100 students per batch)
- Single student add with manual data entry
- Student activation flow via email and SMS with assigned portal login credentials
- Assign students to programs and semesters
- Review and approve pre-recorded AI video modules
- Review and approve AI-generated MCQs
- Create and assign interview sessions for student cohorts (B2B Interview module)
- View and download student-level and cohort-level reports
- RBAC enforcement: Faculty role cannot access institution-wide dashboard or data of other departments

6.4 Student Portal

6.4.1 Classroom — AI Avatar Video Lectures

- Students view AI driven video lectures for their assigned subjects and chapters
- Full chapter video playback with progress tracking.
- Content is displayed in the context of the academic hierarchy (Program > Semester > Subject > Chapter)

6.4.2 QnA AI Tutor

- AI tutor available for doubt resolution after video sessions.
- Interaction supported via voice.
- AI tutor uses the chapter knowledge base for contextually accurate responses.
- Interruption handling: student can interrupt mid-response to ask a follow-up

- Multilingual support: English for primary instruction; Tamil language support for QnA interactions.
- Session context maintained throughout the QnA session

6.4.3 Assessments

- MCQ assessments generated from approved question bank
- Students attempt assessments after completing chapter content
- Auto-evaluation on submission with immediate score display
- Assessment PDF report generated and made available for download post-submission

6.4.4 Leaderboard

- Leaderboard ranking updated in real time after each assessment submission
- Ranking reflects correct scores across all students in the same cohort
- Students can view their position relative to peers

6.4.5 AI Interview Module (B2B Embedded)

- Interview tab embedded within the Student Portal
- Students take AI mock interview sessions with voice interaction and avatar interviewer
- Interview assigned by Faculty/HOD for specific cohorts OR self-initiated by student.
- Interview score report with per-question feedback and personalised improvement plan available post-session.
- For Complete Interview Module separate SOW is maintained.

6.5 Analytics Dashboards

6.5.1 Super Admin Dashboard

- Total institutions onboarded, active students, overall completion rates
- Average assessment scores across all institutions
- Top QnA queries across all institutions
- Content engagement metrics: watch time
- Platform health monitoring

6.5.2 College Admin Dashboard

- Course completion rates by department, subject, and individual student
- Assessment score distributions with pass/fail ratios
- QnA analytics: top 10 most-asked topics per subject
- Interview cohort analytics (B2B): per-student readiness scores, completion rates, weak areas, at-risk interview flags
- Downloadable reports at student and cohort level

7. Notifications and Communication

- Student activation via email and Whatsapp on account creation
- Interview session results shared via email post-completion
- System notifications for content publication, assessment availability, and assignment creation
- Admin notifications on content generation pipeline completion

8. Data and Security

- Role-Based Access Control (RBAC) enforced across all three portal types
- OAuth2 / JWT for authentication and session management
- Data isolation enforced per institution , no cross-institution data access

9. AI Guardrails

The AI system across all modules (QnA Tutor, Interview Engine, Course Generation) shall restrict generation of offensive, unsafe, or inappropriate content

The system shall ensure all AI responses remain within the defined academic or interview context at all times.

10. Technology Stack

Layer	Technology
Frontend (Web)	ReactJS
Backend	Python FastAPI
Database	PostgreSQL
Vector Database	Pinecone / Weaviate
AI / LLM	External LLM API providers: OpenAI, Anthropic, Google
AI Avatar Engine	HeyGen, Synthesia, D-ID, or equivalent avatar generation framework
Speech-to-Text (STT)	Whisper, Deepgram, or equivalent
Text-to-Speech (TTS)	ElevenLabs, Deepgram, or equivalent
Real-Time Infrastructure	LiveKit / Vapi (as required)

Authentication	OAuth2 / JWT
CI/CD	GitHub Actions
Cloud Hosting	AWS or GCP (cloud-native)
File Storage	AWS S3 (session recordings, uploaded content)

11. Phase 1 Milestones and Timeline

Total estimated duration: 6.5 months (26 weeks)

#	Milestone	Duration
M1	Core Platform Foundation & Infrastructure Setup	1 week
M2	Content & Admin Portals	4 weeks
M3	Core Interview Engine	5 weeks
M4	Scoring Intelligence Engine	3 weeks
M5	AI Course Generation Pipeline	5 weeks
M6	Student Portal	5 weeks
M7	Dashboards	2 weeks
M8	UAT, Refinements & Launch	1 week

12. Performance Requirements

- Scalable architecture to support concurrent student users across multiple institutions
- Asynchronous processing for AI content generation pipeline.

13. Terms and Conditions

1. The design will be done based on client references provided during the UI/UX discussion.
2. Server costs, domain, hosting, SSL certificates, professional email, GPU instance shall be borne by the Client. All third-party API costs incurred during and after development shall be borne by the Client.

3. Any additional work outside the agreed scope will be treated as a Change Request (CR) and charged separately based on the efforts required.
4. The total project duration is estimated based on the defined scope. Any additions to scope will result in a revised timeline communicated via a Change Order.
5. A 12-month software warranty is provided from the date of production deployment. Warranty terms to be detailed in a separate Warranty Schedule.
6. Support and Maintenance post-launch will be governed by a separate Support SLA document to be agreed prior to go-live.

14. Exclusions

The following are explicitly out of scope for Phase 1 and will be addressed in future phases or separate engagements:

- Mobile application (iOS / Android)
- Offline access or downloaded content
- Live video classes or real-time faculty-to-student streaming
- B2C standalone Interview Portal subscription billing (covered in separate B2C SOW)
- On-premise deployment (available as an add-on at additional cost)
- Integration with third-party LMS platforms (e.g., Moodle, Canvas)

15. Acceptance

This Statement of Work is accepted and agreed to by both parties.

For Knotopian

Signature: _____

Name: _____

Title: _____

Date: _____

For Rytsense Technologies

Signature: 

Name: RAMKUMAR P

Title: CEO & Co-Founder

Date: 01/04/26